AIoT Prototype Development & Evaluation Report: On-Device Multi-Modal Bearing Fault Detector

File: BEARING-FAULT-DETECTOR-REPORT.pdf | **Target MCU:** Arduino Nano 33 BLE Sense Rev2 (nRF52840, 64 MHz Cortex-M4F, 256 KB SRAM, 1 MB Flash)

1. Prototype Objectives, Problem Statement, and Chapter 1.6 Evaluation

Rolling-element bearings account for over 40% of unexpected failures in industrial rotating machinery. Traditional cloud-centric Condition-Based Maintenance (CBM) pipelines stream raw high-frequency vibration (1.6 kHz) and ultrasonic acoustic (16 kHz) waveforms continuously over Wi-Fi, creating severe bandwidth congestion, latency jitter, and battery depletion within days. This prototype solves that problem by moving the entire Digital Signal Processing (DSP) and Deep Learning classification pipeline onto an ultra-low-power microcontroller node (Arduino Nano 33 BLE Sense Rev2). Evaluating the architecture under **Course Chapter 1.6 (Edge vs. Cloud Trade-offs in AIoT)**: (1) **Latency Determinism:** On-device inference executes in a bounded 24.8 ms window, eliminating network round-trip variability; (2) **Bandwidth & Energy Efficiency:** Instead of streaming 35.2 KB/s of raw waveforms, the node transmits a sparse 12-byte BLE 5.0 GATT alert packet only when an anomaly is classified with >=85% confidence, reducing wireless duty cycle by >99.8%; (3) **Privacy & Reliability:** Acoustic streams never leave the sensor node, and fault detection continues autonomously during factory network outages.

1.1 Iterative Development Process & Abandoned Approaches

The prototype evolved across four engineering iterations: **Iteration 1 (Abandoned Raw Wi-Fi Streaming on ESP32):** Streaming raw 16 kHz audio over Wi-Fi drew 168 mA average current (554.4 mW), exhausting a 1,000 mAh LiPo battery in 5.9 hours and suffering 14% UDP packet loss under industrial RF interference. **Iteration 2 (Abandoned Raw Time-Domain 1D-CNN):** Feeding 2,048-sample raw waveforms into a 5-layer 1D-CNN required a 192 KB activation buffer and 318 KB Flash weights, triggering an Out-Of-Memory (OOM) fault on the 256 KB SRAM budget. **Iteration 3 (Multi-Modal Spectral Feature Fusion):** Extracting 32x32x2 dual-channel short-time spectral frames (vibration + acoustic envelope) plus thermal drift reduced input dimensionality by 4x while explicitly isolating Bearing Pass Frequency Outer/Inner (BPFO/BPFI) harmonics. **Iteration 4 (Structured Pruning + INT8 PTQ):** Applying 50% structured magnitude pruning and full integer quantization yielded a 26.4 KB model that runs in 24.8 ms inside a 36.8 KB tensor arena.

2. Sensing Pipeline: Hardware, Sensors, Communications, and Energy Budget

The sensing pipeline fuses three physical modalities chosen to capture early-stage micro-spalling (acoustic HF), mid-stage structural impacts (triaxial vibration), and late-stage friction degradation (thermal drift). Rather than continuous CPU polling—which prevents the Cortex-M4F from entering sleep—the pipeline uses a hierarchical event-driven and DMA-buffered reading architecture:

Component / Sensor	Captured Signal & Range	Reading Mode & Interface	Sampling Freq.	Active Current
BMI270 / LSM9DS1 IMU	3-Axis Acceleration (+/-16g, 16-bit)	Event-Based Wake (>0.35g) + 32-sample FIFO Burst (I2C 400kHz)	1.6 kHz	0.95 mA
MP34DT05-A MEMS Mic	Acoustic Pressure (64 dB SNR)	Hardware EasyDMA Double-Buffered Streaming (PDM)	16,000 Hz	0.65 mA
TMP117 Contact Probe	Bearing Housing Temp (+/-0.1 C)	Low-Rate Periodic Polling (I2C 0x48)	0.2 Hz (1/5s)	0.0035 mA
nRF52840 BLE 5.0 Radio	12-Byte GATT Anomaly Alert Packet	Conditional Event-Driven TX (only when Fault Confidence >=85%)	Sparse (<0.1 Hz)	4.8 mA (TX peak)

Energy Budget Justification: In Idle Monitoring (machine off or vibration <0.35g), the MCU stays in System ON Sleep (3.2 uA) while the IMU monitors wake-on-motion at 15 uA. During active machinery operation, acquiring a 320 ms window every 2.0 seconds (16% duty cycle) plus a 24.8 ms inference burst yields an average current of **0.89 mA (2.94 mW @ 3.3 V)**, enabling **46.8 days** of continuous operation on a 1,000 mAh cell (a 189x improvement over the abandoned Wi-Fi streaming prototype).

3. On-Device AI: Architecture, Optimization Hyperparameters & Pareto Frontier

The classifier is a 2D Depthwise-Separable Convolutional Neural Network (DS-CNN) accepting a [1, 32, 32, 2] spectral input tensor and outputting probabilities across 4 classes: Normal (0), Inner Race Fault (1), Outer Race Fault (2), and Ball Defect (3). To fit strictly within the nRF52840 memory hierarchy and maximize CMSIS-NN SIMD acceleration

(`__SMLAD`), two compound optimizations were applied with the following exact parameters:

1. Structured Magnitude Pruning (`tfmot.sparsity.keras.prune_low_magnitude`): Schedule = `PolynomialDecay(initial_sparsity=0.0, final_sparsity=0.50, begin_step=1000, end_step=5000, frequency=100, power=3)`. Pruning was applied to all depthwise/pointwise Conv2D and Dense weight tensors prior to fine-tuning for 15 epochs (Adam, lr=5e-4).
2. Full Integer Post-Training Quantization (INT8 PTQ): Converter configured with `optimizations = [tf.lite.Optimize.DEFAULT]`, `supported_ops = [tf.lite.OpsSet.TFLITE_BUILTINS_INT8]`, and `inference_input_type = inference_output_type = tf.int8`. Calibrated using 200 stratified spectral frames, producing input scale S = 0.03125 (2^-5) and zero-point Z = -12.

Optimization Stage	Precision	Sparsity	Test Acc.	Macro F1	Flash Size	Arena RAM	Latency	Pareto?
1. Baseline DS-CNN	FP32	0%	96.4%	0.963	148.6 KB	94.2 KB	112.4 ms	No (Baseline)
2. Dynamic Range Quant.	INT8 W / FP32 A	0%	96.1%	0.960	44.8 KB	78.5 KB	68.2 ms	Yes
3. Pruned 30% + INT8 PTQ	Full INT8	30%	95.8%	0.957	34.2 KB	41.0 KB	34.6 ms	Yes
4. Pruned 50% + INT8 PTQ*	Full INT8	50%	95.3%	0.952	26.4 KB	36.8 KB	24.8 ms	YES (DEPLOYED)
5. Pruned 75% + INT8 PTQ	Full INT8	75%	86.2%	0.854	18.1 KB	32.4 KB	19.5 ms	No (Collapsed)

Pareto Frontier Analysis: Stage 4 (50% Pruned + Full INT8 PTQ) achieves the knee of the Pareto frontier: compared to the FP32 baseline, it reduces Flash footprint by **5.63x** (148.6 KB to 26.4 KB), reduces peak SRAM tensor arena by **2.56x** (94.2 KB to 36.8 KB), and accelerates inference latency by **4.53x** (112.4 ms to 24.8 ms) at the cost of only 1.1% accuracy degradation. Pushing sparsity to 75% (Stage 5) removes critical BPFO/BSF harmonic filters, causing a steep 9.1% accuracy collapse.

4. Self-Reflection, Course Concept Application & AI Tool Disclosure

Prototype Limitations & Future Work: (1) *Variable RPM Domain Shift:* Because characteristic fault frequencies scale linearly with shaft speed, operating the motor outside the trained 1,750-1,800 RPM band shifts spectral peaks across bins; future iterations will integrate a Hall-effect tachometer for order-tracked resampling. (2) *Static Threshold Calibration:* On-device TinyOL (Tiny Online Learning) on the final dense layer will be added to self-calibrate baseline vibration on newly installed machines.
Course Concept Applied — Hardware-Aware Co-Design & Cascade Wake-Up Architecture: Rather than treating sensor acquisition and neural network design as separate silos, hardware-aware co-design aligns sensor FIFO depths, fixed-point FFT dimensions (32x32), and INT8 SIMD tensor layouts (`alignas(16)`) so that expensive compute stages are gated behind ultra-low-power hardware comparators.
Evolution of AIoT Understanding: Completing this project transformed my perspective of AIoT from 'running compressed cloud models on microcontrollers' to holistic system engineering where energy per bit sensed, processed, and transmitted dictates every architectural boundary.

5. Mandatory Declaration of AI Tool Usage

In accordance with course academic integrity guidelines, the use of AI tools during the prototype development is declared below:
- **AI Tool Used:** Google Gemini
- **Model / Version Used:** Gemini (2026 release)
- **Purpose of Use:** Assisting with debugging TensorFlow Lite for Microcontrollers (TFLM) C++ tensor arena alignment errors, verifying CMSIS-NN INT8 quantization scale/zero-point arithmetic, and structuring repository documentation for reproducibility.